Description

Illustration





Figure caption

1. A single block of autoclaved aerated concrete. 2. Aerated concrete blocks within the walls of a building under construction. Images used under license from Shutterstock.com

The material

Autoclaved aerated concrete (AAC) is a foamed or cellular concrete produced by adding a foaming agent to the concrete mix in order to entrap air bubbles during the curing process, which involves steam heating ("autoclaving"). The resultant low density product is used to make lightweight buildings, structures and decorations. The strength of the concrete can be improved by the addition of fibers, but the values quoted here are for unreinforced concrete.

Compositional summary

0.6:1:0:4 Water:OPC:Fine:Coarse
Mix constituents and proportions are varied to produce widely ranging properties.

General properties

Density	400	-	900	kg/m^3
Price	* 0.0453	-	0.0604	GBP/kg
Date first used	1929			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	✗

Mechanical properties

Young's modulus		12	-	18	GPa
Shear modulus	*	5.1	-	7.6	GPa
Bulk modulus	*	6.3	-	9.5	GPa
Bending modulus	*	12	-	18	GPa
Poisson's ratio		0.17	-	0.2	
Yield strength (elastic limit)	*	0.6	-	1.1	MPa
Tensile strength	*	0.6	-	1.1	MPa
Compressive strength	*	1.2	-	1.87	MPa
Bending strength		0.7	-	1.3	MPa
Elongation	*	0	-	0.01	% strain
Hardness - Vickers	*	0.36	-	0.56	HV
Fatigue strength at 10^7 cycles	*	0.5	-	0.9	MPa
Fracture toughness	*	0.04	-	0.06	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		1.25	-	1.43	m.°C/W
Thermal expansion coefficient		7	-	12	µstrain/°C
Specific heat capacity	*	860	-	1.1e3	J/kg.°C
Melting point		930	-	1.2e3	°C
Maximum service temperature		200	-	300	°C
Minimum service temperature		-160			°C
Flammability		Non-flammable			
Emissivity		0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability		6.01e-12	-	2.8e-11	kg.m/s.m^2.Pa
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity		3.16e11	-	3.16e12	µohm.cm
Dielectric constant (relative permittivity)	*	8	-	12	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	0.5	-	2	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Good
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4	
Machinability	4	-	5	

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.425	-	0.469	kg/kg
Embodied energy (virgin grade production)	* 3.23	-	3.57	MJ/kg
Water Usage	* 3.34	-	3.69	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 15.5	-	16.8	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.01	-	1.1	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0425	-	0.0469	kg/kg
Embodied energy, recycling	* 0.452	-	0.5	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Autoclaved Aerated Concrete (AAC) is light colored and contains many small air bubbles that contribute to the material's insulating properties. AAC can be sculpted with wood-working tools but, being soft, it is rarely used in an exposed external condition, and so requires surface protection. AAC has potential as an environmentally responsible building material.

Reinforced elements can be produced but owing to the low durability of steel in contact with air, stainless or epoxy-coated bar has to be used to prevent corrosion.

Technical notes

AAC has low compressive strength, but sufficient for it to be used as a load-bearing material in structures up to 3 stories high. Entire buildings can be made in AAC including walls, floors and roofing. Lintels, floor slabs and roofing panels require steel reinforcement to give structural adequacy. AAC has reasonably good insulation qualities and its low thermal mass gives energy efficiency. It provides good sound insulation and is suited for fire-rated applications.

Because it is porous, it will absorb water (e.g. if exposed to rain). There are a number of proprietary finishes based on acrylic polymers, which, when mixed with sand and cement, provide a durable and water-resistant coating.

The quoted properties are for unreinforced AAC. However, in practice, they are significantly improved by the addition of reinforcement as bar or fibers.

Typical uses

Concrete blocks, lightweight tilt-up panels, foamed concrete floor screeds, sound and thermal insulation, geotechnical and ornamental concrete applications - e.g. column heads

Tradenames

LiteBuilt, Hebel, Ytong

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1686: Standard Practice for Installation and Testing of Reinforced Autoclaved Aerated Concrete (AAC) Units
ASTM C1692: Standard Practice for Construction and Testing of Autoclaved Aerated Concrete (AAC) Masonry
ASTM C1693: Standard Specification for Autoclaved Aerated Concrete (AAC)
ASTM C1694: Standard Specification for Reinforced Autoclaved Aerated Concrete (AAC) Elements
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
BS 8500: Concrete. Complementary British Standard to BS EN 206
EN 12602: Prefabricated reinforced components of autoclaved aerated concrete
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity

Further reading

The Australian Government web site www.greenhouse.gov.au carries useful information on AAC.

Links

Producers

Reference

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.